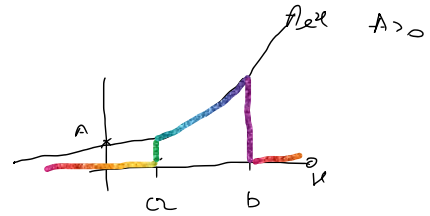


$$a < b$$

$$\int_a^b e^x dx = [e^x]_a^b = e^b - e^a$$



$$f_X(x) = \begin{cases} A e^x & x \in (a, b) \\ 0 & x \notin (a, b) \end{cases} \rightarrow A = \frac{1}{e^b - e^a}$$

$$\int_a^b f_X(x) dx = \int_a^b A e^x dx = A [e^b - e^a] = 1$$

$$A = \frac{1}{e^b - e^a}$$

$a < b \Rightarrow > 0$

$X$   
 $| X > c$

$$c < a \rightarrow P[X > c] = 1$$

$$c \in (a, b) \rightarrow 0 < P[X > c] < 1$$

$$c > b \rightarrow P[X > c] = 0$$

$$b - \varepsilon > c > b \xrightarrow{\varepsilon \rightarrow 0} 0$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \frac{P(B)}{P(B)} = 1$$

$$P[X > c | X = u] = \frac{P[X > c \cap X = u]}{P[X = u]} = \frac{P[X = u]}{P[X = u]} = 1$$

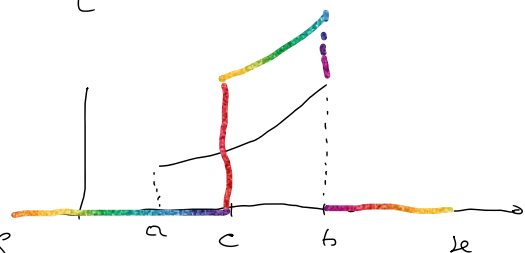
$$f_{X|X>c}(u) = \frac{P[X > c | X = u]}{P[X > c]} f_X(u) = \begin{cases} 0 & u \leq c \\ \frac{e^b - e^c}{e^b - e^a} f_X(u) & c < u < b \\ 0 & u > b \end{cases}$$

$$c \in (a, b)$$

$$P[X > c] = \int_c^b f_X(u) du = \int_c^b A e^u du = A [e^b - e^c] = \frac{e^b - e^c}{e^b - e^a}$$

$$P[X > c | X = u] = P[X > c] = \begin{cases} 1 & u > c \\ 0 & u \leq c \end{cases}$$

$$f_X(x) = \begin{cases} 0 & x \leq c \\ \frac{e^b - e^c}{e^b - e^a} \frac{1}{e^b - e^c} e^x & c < x < b \\ 0 & x > b \end{cases} = \begin{cases} \frac{1}{e^b - e^c} e^x & c < x < b \\ 0 & x \in (c, b) \end{cases}$$



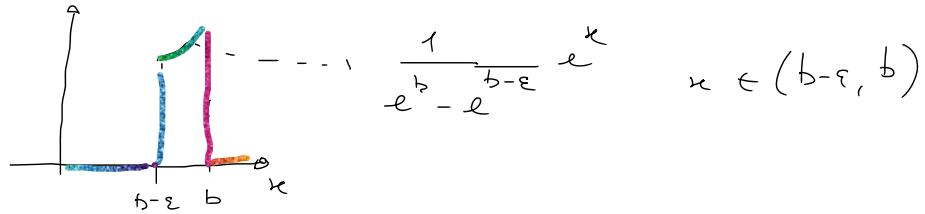
$$c < a$$

$$P[X > c] = \begin{cases} 1 & c < a \\ 0 & c > b \end{cases}$$

$$c < a$$

$$P[X > c | X = x] = P[X > c] = \begin{cases} 1 & x > c \\ 0 & x \leq c \end{cases} \quad P[X > c] = 1 \quad \forall \quad x \in (a, b)$$

$$c = b - \varepsilon$$



$$\left. \begin{aligned} X &\sim \mathcal{N}(\mu_x, \sigma_x^2) \\ Y &\sim \mathcal{N}(\mu_y, \sigma_y^2) \end{aligned} \right\} \begin{aligned} Z &= aX + bY \\ \rho &= \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} \end{aligned} \rightarrow f_Z(z) = \frac{1}{\sqrt{2\pi}\sigma_z} e^{-\frac{(z-\mu_z)^2}{2\sigma_z^2}}$$

$$\mu_z = E[Z] = E[aX + bY] = aE[X] + bE[Y] = a\mu_x + b\mu_y$$

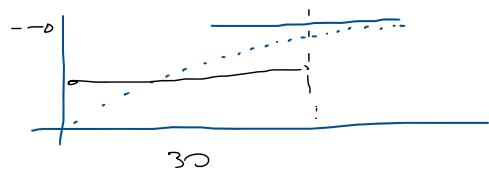
$$E[Z^2] = E[(aX + bY)^2] = E[a^2X^2 + b^2Y^2 + 2abXY] = a^2E[X^2] + b^2E[Y^2] + 2abE[XY]$$

$$\sigma_z^2 = E[Z^2] - \mu_z^2 = a^2E[X^2] + b^2E[Y^2] + 2abE[XY] - (a\mu_x + b\mu_y)^2$$

$$= a^2[E[X^2] - \mu_x^2] + b^2[E[Y^2] - \mu_y^2] + 2ab(E[XY] - \mu_x\mu_y)$$

$$= a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\text{cov}[X, Y]$$

$$= a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_x\sigma_y\rho$$



$$X_i \in \{x_1, x_2, \dots, x_{|X|}\}$$

$$P[X_1 = x_{i_1}] \quad i_1 \in \{1, 2, \dots, |X|\}$$

$$P[X_1 = x_{i_1}, X_2 = x_{i_2}]$$

$$i_1, i_2 \in \{1, 2, \dots, |X|\} = \{1, 2, \dots, |X|\}$$

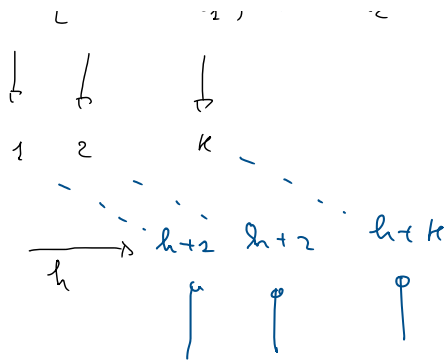
$$(i_1, i_2) \in \{1, 2, \dots, |X|\}^2 = \{1, 2, \dots, |X|\}^2$$

$$P[X_1 = x_{i_1}, X_2 = x_{i_2}, \dots, X_k = x_{i_k}]$$

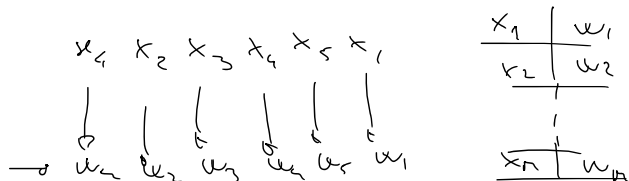
$$i_1, i_2, \dots, i_k \in \{1, \dots, |X|\} = \{1, 2, \dots, |X|\}$$

$$(i_1, i_2, \dots, i_k) \in \{1, \dots, |X|\}^k = \{1, 2, \dots, |X|\}^k$$

$$\downarrow \quad \downarrow \quad \downarrow$$



$(i_1, i_2, \dots, i_h) \in \{1, \dots, n\}^h = \{1, 2, \dots, |X|\}^h$   
*non dipende propriamente da h*



$$\frac{2^k}{2}$$

|           |            |               |
|-----------|------------|---------------|
| $x_1$     | $x_1, x_1$ | $w_1$         |
| $x_2$     | $x_1, x_2$ | $w_2$         |
|           | ...        | ...           |
|           | $x_1, x_n$ | $w_n$         |
|           | $x_2, x_1$ | $w_{n+1}$     |
|           | ...        | ...           |
|           | $x_2, x_n$ | $w_{2n}$      |
|           | ...        | ...           |
|           | $x_n, x_1$ | $w_{(n-1)+1}$ |
|           | ...        | ...           |
| $x_{n^2}$ | $x_n, x_n$ | $w_{n^2}$     |

$$H(x) < L = \underline{E[\ell(x)]} < H(x) + 1$$

$$E[\ell(x_i, x_{i-1})] = E[\ell(x^2)]$$

$$H(x^2) < \underline{E[\ell(x^2)]} < H(x^2) + 1$$

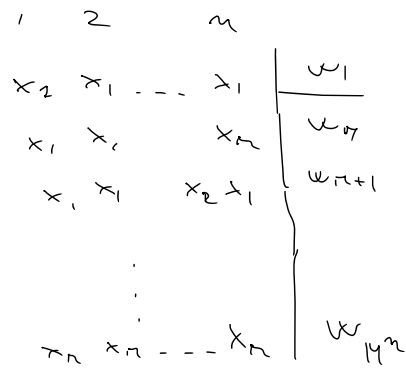
$$\frac{H(x^2)}{2} < \frac{E[\ell(x^2)]}{2} < \frac{H(x^2) + 1}{2}$$

|                 |                  |
|-----------------|------------------|
| $x_1, x_1, x_1$ | $w_1$            |
| ...             | ...              |
| $x_1, x_1, x_n$ | $w_n$            |
| $x_1, x_2, x_1$ | $w_{n+1}$        |
| ...             | ...              |
| $x_1, x_2, x_n$ | $w_{2n}$         |
| $x_1, x_n, x_1$ | $w_{(n-1)+1}$    |
| ...             | ...              |
| $x_1, x_n, x_n$ | $w_{n^2}$        |
| $x_2, x_1, x_1$ | $w_{n^2+1}$      |
| ...             | ...              |
| $x_2, x_n, x_n$ | $w_{2n^2}$       |
| $x_3, x_1, x_1$ | $w_{2n^2+1}$     |
| ...             | ...              |
| $x_3, x_n, x_n$ | $w_{3n^2}$       |
| ...             | ...              |
| $x_n, x_1, x_1$ | $w_{(n-1)n^2+1}$ |

$$x_i, x_{i+1}, x_{i+2} = x^3 \quad E[\ell(x^3)]$$

$$H(x^3) < E[\ell(x^3)] < H(x^3) + 1$$

$$\frac{H(x^3)}{3} < \frac{E[\ell(x^3)]}{3} < \frac{H(x^3) + 1}{3}$$



$$E[\ell(x^3)]$$

$$(x_i, x_{i+1}, x_{i+2}, \dots, x_{i+n-1}) = x^n$$

$$\begin{matrix} x_1, x_2, x_3 \\ \vdots \\ x_n, x_{n+1}, x_{n+2} \end{matrix} \left| \begin{matrix} w_{(n-1)n+1} \\ \vdots \\ w_{n+1} \end{matrix} \right.$$

$$(x_1, x_{i+1}, x_{i+2}, \dots, x_{i+n-1}) \equiv \underline{x}^n$$

$$H(\underline{x}^n) < \mathbb{E}[L(\underline{x}^n)] < H(\underline{x}^n) + 1$$

$$\frac{H(\underline{x}^n)}{n} < \frac{\mathbb{E}[L(\underline{x}^n)]}{n} < \frac{H(\underline{x}^n)}{n} + \frac{1}{n}$$

$$P[X(n) = x_n | X(n-1) = x_{n-1}]$$

$$X = \{x_1, x_2, \dots, x_n\} \quad |X| = n$$

$$X = \{1, 2, \dots, M\} \quad |X| = M$$

$$P[X(n) = x_n | X(n-1) = x_{n-1}]$$

$$x_n \in \{1, \dots, M\}$$

$$x_{n-1} \in \{1, \dots, M\}$$

$$\begin{matrix} x_n \\ \vdots \\ x_{n-1} \end{matrix} = \underline{x}_n$$

$$P[X(n) = x_n | X(n-1) = x_{n-1}, X(n-2) = x_{n-2}, X(n-3) = x_{n-3}, \dots]$$

$$x_{n-2}, x_{n-3}, x_{n-4} \in \{1, \dots, M\}$$

$$p_n = P[X(n) = x_n] \quad n \in \{1, \dots, M\}$$

$$\{p_1, p_2, \dots, p_M\}$$

$$\text{Indip. del processo} \Rightarrow P_i^{(j)} = p_i \quad \forall j$$

$$P_i^{(j)} = P[X(n) = i | X(n-1) = j] \quad i, j \in \{1, \dots, M\}$$

$$P[X(n) = i | X(n-1) = j, X(n-2) = k] \quad i, j, k \in \{1, \dots, M\}$$

$$P_i^{(j,k)}$$

$$\text{Consider. indep. de } X(n-2) \text{ e } X(n-1) \Rightarrow P_i^{(j)} = P_i^{(j,k)} \quad \forall k \in \{1, \dots, M\}$$

$$P[X(n) = i | X(n-1) = j, X(n-2) = k, X(n-3) = l] \quad i, j, k, l \in \{1, \dots, M\}$$

$$P_i^{(j,k,l)}$$

$$P_i^{(j,k,l)} \Rightarrow \sum_{l=1}^M P_i^{(j,k,l)} = 1 \quad \forall j, k \in \{1, \dots, M\}$$

$$P_i^{(j,k,l)} = P_i^{(j)} \quad \forall k, l \in \{1, \dots, M\}$$

$$X(n) \text{ \u00e9 condic. indep. de } X(n-2) \text{ e } X(n-3) \text{ de } X(n-1)$$

$$P[X(n)=i \mid X(n-1)=j, X(n-2)=k, X(n-3)=l, X(n-4)=m]$$

$$P_i^{(j,k,l,m)}$$

$$P_i^{(j,k,l,m)}$$

$$\geq 0 \quad \forall i, j, k, l, m$$

$$\sum_{i=1}^N P_i^{(j,k,l,m)} = 1 \quad \forall j, k, l, m$$

$$P_i^{(j,k,l,m)} = P_i^{(j)} \quad \forall k, l, m \in \{1, \dots, N\}$$

$X(n)$  è condit. indep. da  $X(n-2), X(n-3), \dots, X(n-4)$  dato  $X(n-1)$

Andrej Andrejivich Markov

$$P = \begin{bmatrix} P_1^{(1)} & P_2^{(1)} & \dots & P_N^{(1)} \\ P_1^{(2)} & P_2^{(2)} & \dots & P_N^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ P_1^{(n)} & P_2^{(n)} & \dots & P_N^{(n)} \end{bmatrix}$$

$$P_i$$

$$P_i^{(j)}$$